CHAPTER 8

Hierarchical Clustering and Dendrogram

You may wonder why I do not explain hierarchical clustering right after k-means clustering. Well, there are some reasons. The first one is that Excel is not as suitable for hierarchical clustering or dendrogram as it is for k-means clustering. I would like to introduce a few more other popular data mining methods before diving into this challenging topic. The second reason is that I consider it suitable to explain hierarchical clustering right after K-NN. The last reason is because of dendrogram. To draw a dendrogram in Excel requires quite some manual work, which could be boring to some learners.

Please download the sample Excel files from https://github.com/hhohho/Learn-Data-Mining-through-Excel-2 for this chapter's exercises.

General Understanding

Hierarchical clustering is similar to k-means clustering in that it is also an unsupervised classification method that categorizes subjects (data points) into different groups (clusters) based on the similarity of certain characteristics. However, these two clustering methods are based on two different algorithms, which results in different final products. K-means clustering generates parallel clusters. Hierarchical clustering produces a tree of clusters as its name indicates, instead. Moreover, the final product of hierarchical clustering is usually presented in the form of a dendrogram that shows the relative distances of the original data points or grouped data points. This is the major reason why hierarchical clustering is frequently used in biomedical sciences to show the clustering of genes or samples.

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K-means clustering starts with defining the number k, the number of clusters. Hierarchical clustering starts with a proximity matrix and never needs to predefine the number of clusters. The algorithm of hierarchical clustering can be explained by the following example that contains six data points A, B, C, D, E, and F.

1. Every data point is treated as a cluster initially.
2. A proximity matrix is constructed to show the distances among the six clusters.
3. Assume that the smallest distance is between A and C, then A and C are placed in the same cluster in the first round and they are represented as AC.
4. Now, there are five clusters only: AC, B, D, E, and F.
5. Repeat step 2 to create a proximity matrix for the five clusters. Assume that B and D have the least distance, then there are only four clusters: AC, BD, E, and F.
6. Repeat step 2 to create a proximity matrix for the four clusters. Assume that AC and F have the least distance, then there are only three clusters: ACF, BD, and E.
7. Repeat step 2 to create a proximity matrix for the three clusters. Assume that ACF and BD have the least distance, then there are only two clusters: ACFBD and E.
8. Hierarchical clustering stops at only one cluster: ACFBDE.

The preceding algorithm is for agglomerative clustering, a bottom-up hierarchical clustering. The opposite of agglomerative clustering is called divisive clustering, which is top down. In this book, we will experience agglomerative clustering, and the distance among clusters is again the Euclidean distance.

What is dendrogram? Well, a graph truly worth more than one thousand words in this case. Following the preceding example, the final product of the algorithm can be represented by the following diagram which is a bona fide dendrogram.

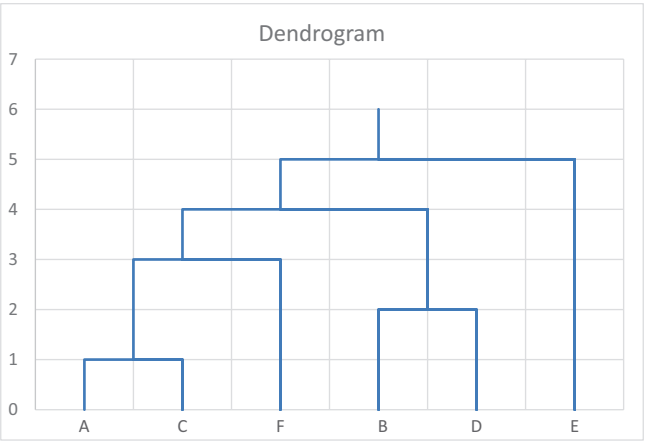


Figure 8-1. *A sample dendrogram showing the hierarchical clustering process of A,B,C,D,E,F*

Observe in Figure 8-1 that (1) there are five levels for six data points, one level for one round of clustering and (2) A and C are first clustered together, and therefore the clade for AC has the least height. In dendrogram, the higher the clade, the larger the distance between the two clusters.

Draw a Dendrogram in Excel Without Add-ins

Before I show you how to use Excel to implement the agglomerative hierarchical clustering algorithm, let's learn how to draw a dendrogram first in Excel. Note, there are commercial analytical software that can be purchased and added into Excel to ease the hierarchical clustering process and dendrogram creation. What I am showing here does not require any add-ins.

Let's start from the simplest dendrogram as shown in Figure 8-2.

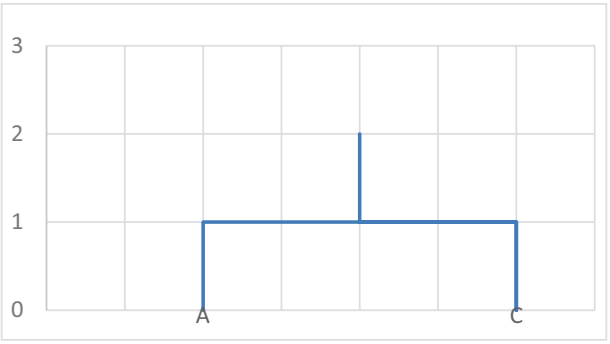


Figure 8-2. *The simplest dendrogram*

If I ask you to use a pen to draw the chart in Figure 8-2 on a piece of paper under the condition that your pen can never leave the paper until completion, how are you going to draw it? Well, you can start from the top, right, or left. Assume you start from the left where the letter A stays (height=0), your pen should move as follows:

1. Start from A, go up by 1.
2. Turn to right, stop at above letter C.
3. Go down by 1 to letter C.
4. Go up by 1.
5. Turn left and stop at the middle between A and C.
6. Go up by 1 to 2.

What if the dendrogram looks like Figure 8-3 and we need to stop at height 3?



Figure 8-3. *A slightly more complicated dendrogram*

Well, we can continue moving our pen as follows:

7. Turn our pen to right so that it is right above letter F.
8. Go down by 2 to 0.
9. Go up by 2.
10. Turn left and stop above letter C.
11. Go up by 1 to 3.

So far, we should have understood that our pen goes through some paths twice. Yes, that is the trick. If we convert these pen movements into a scatter plot in Excel, then the movements can be represented by the data points shown in Figure 8-4.

	A	B	C	D	E	F	G	H
1	X	Y						
2	1	0		Start from (1, 0)				
3	1	1		Go up by 1 (1, 1)				
4	3	1		Go right by 2 (3, 1)				
5	3	0		Go down by 1 (3, 0)				
6	3	1		Go up by 1. Here, go back to (3, 1)				
7	2	1		Go left by 1. Here, go back to (2, 1)				
8	2	2		Go up by 1 to (2, 2)				
9	5	2		Go right by 3 (5, 2)				
10	5	0		Go down by 2 (5, 0)				
11	5	2		Go up by 2. Here, go back to (5, 2)				
12	3	2		Go left by 2 (3, 2)				
13	3	3		Go up by 1 (3, 3)				

Figure 8-4. *The X and Y values for a scatter plot that produces a dendrogram*

Please open Chapter8-1a.xlsx. What will jump into your eyes should look like Figure 8-5, which is a dendrogram slightly different from Figure 8-3 because it does not have the labels A, C, or F. Note, this dendrogram is a scatter plot and is produced based on the data in A1:B13 which is the same as shown in Figure 8-4.

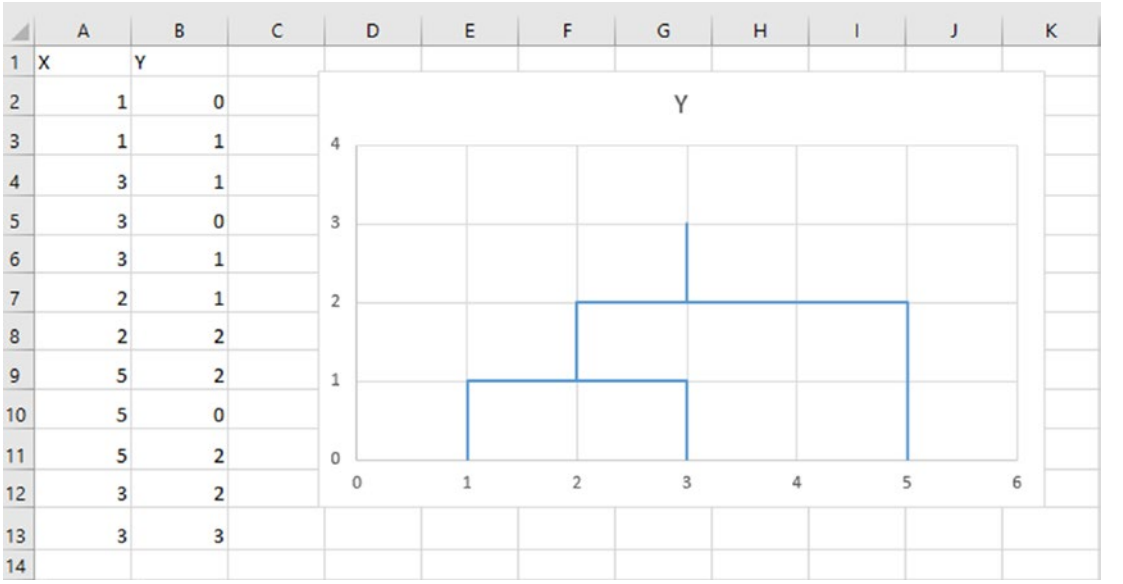


Figure 8-5. *The dendrogram without labels*

To make the dendrogram in Figure 8-5 look like that in Figure 8-3, we need to complete the following:

12. As shown in Figure 8-6, click inside the x-axis, then select Delete on the upcoming menu. This action deletes the x-axis.

13. Right-click on the chart title to delete it, too, as shown in Figure 8-6.

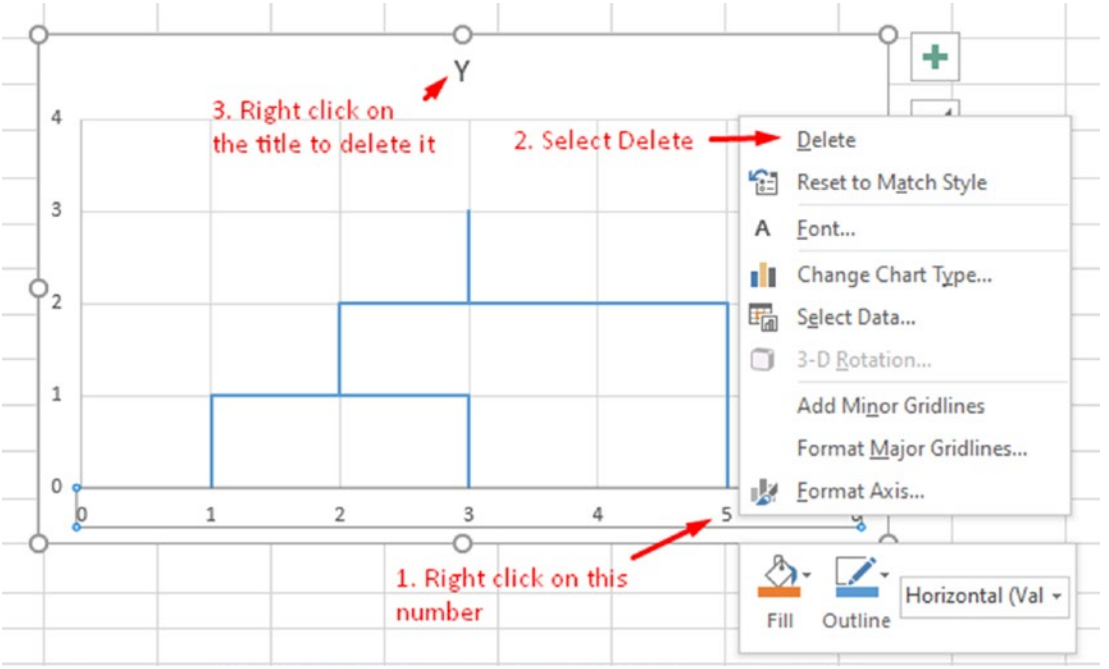


Figure 8-6. *Delete the x-axis and the chart title*

14. Follow Figure 8-7 to add data labels. Make sure to select Below.

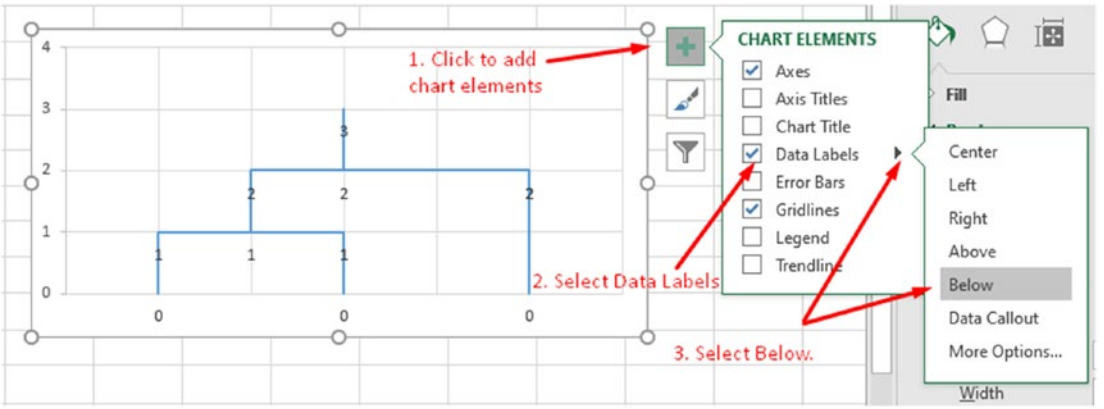


Figure 8-7. *Add data labels*

15. Follow Figure 8-8 to delete data labels one by one but leave the bottom three labels intact. Bear in mind that this action requires some effort.

Note, when you click on one data label, all data labels are selected. You should continue clicking on that specific label (single click). This action will deselect other data labels but only select the one you clicked.

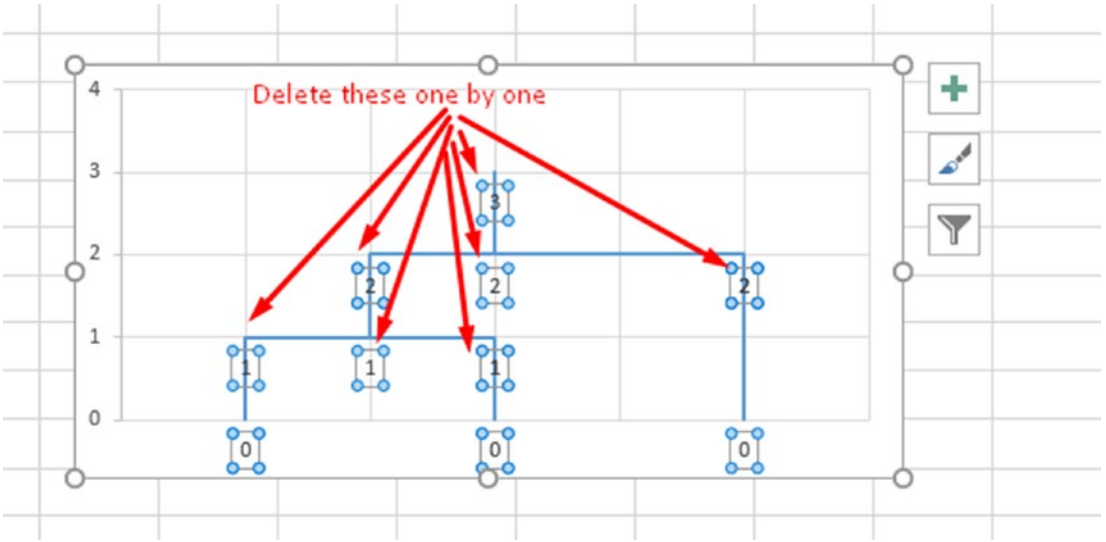


Figure 8-8. *Delete the marked data labels one by one. Leave the bottom three labels intact*

16. For the three bottom data labels, click on the leftmost one. All the three labels are selected. Immediately click on the leftmost data label again; now only the leftmost data label is selected. Change its text to A.

17. Continue to change the text of the rest two data labels. Figure 8-9 shows the text changing for the last data label. After this step, we shall get a dendrogram similar to Figure 8-3.

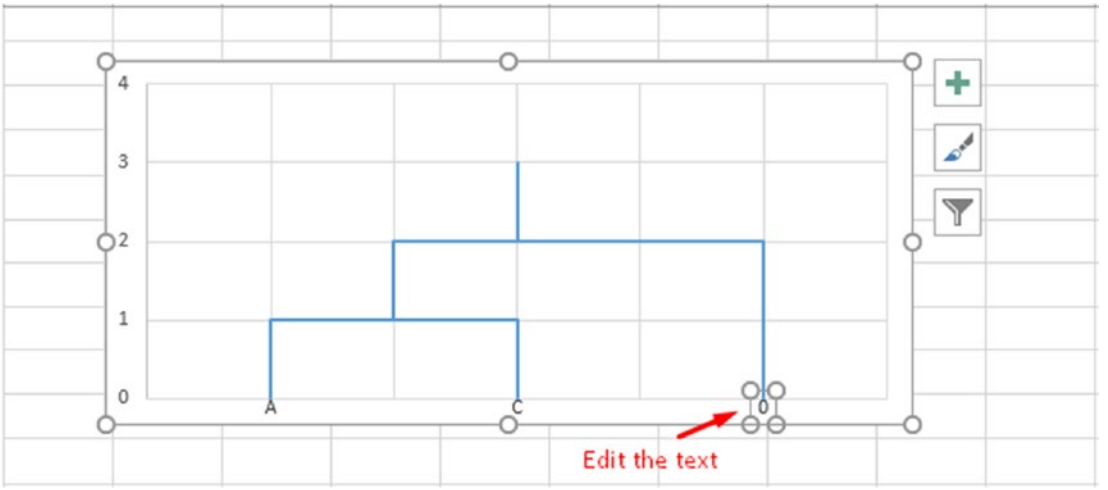


Figure 8-9. *Edit the text inside a data label*

The final result is available in Chapter8-1b.xlsx.

Learn Hierarchical Clustering Through Excel

Let's open the file Chapter8-2a.xlsx which contains ten data records. Using Excel to demonstrate hierarchical clustering with too many records won't be practical because both dendrogram creation and the hierarchical clustering process require quite some manual work. As the purpose of this chapter is to let you experience how hierarchical clustering works, there is no need to start with too many records.

You may have already noticed that these ten records are the first ten records from Chapter3-1a.xlsx. These ten records are for the ten countries L1, L2, …, L10 (column B). Our task is to hierarchically cluster these ten data points and then observe their relative distances or similarities.

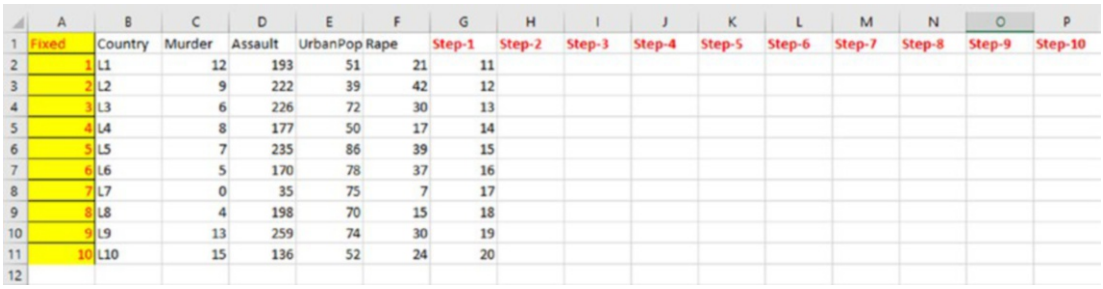
Please follow these instructions to execute the hierarchical clustering algorithm as explained earlier:

1. In cells G1:P1, enter the text words "Step-1", "Step-2", "Step-3", "Step-4", "Step-5", "Step-6", "Step-7", "Step-8", "Step-9", and "Step-10". You can achieve this by entering "Step-1" in cell G1 and then autofill to P1.

2. To help differentiate the original data inside columns C:F from the Steps, it is a good idea to format the text words in G1:P1 such that they are bold and red (or another non-black color).

3. In cells G2:G11, enter numbers 11–20. These numbers represent the current clusters each data point is in. Note, since we have ten data points, any integer greater than ten can mark that this data point hasn't been clustered. This is the initial step, that is, Step-1.

   At this moment, our worksheet should look like Figure 8-10.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Fixed	Country	Murder	Assault	UrbanPop	Rape	Step-1	Step-2	Step-3	Step-4	Step-5	Step-6	Step-7	Step-8	Step-9	Step-10
2	1	L1	12	193	51	21	11									
3	2	L2	9	222	39	42	12									
4	3	L3	6	226	72	30	13									
5	4	L4	8	177	50	17	14									
6	5	L5	7	235	86	39	15									
7	6	L6	5	170	78	37	16									
8	7	L7	0	35	75	7	17									
9	8	L8	4	198	70	15	18									
10	9	L9	13	259	74	30	19									
11	10	L10	15	136	52	24	20									
12																

Figure 8-10. *The initial step of hierarchical clustering*

4. In cells B13:K13, enter integers 1–10, respectively. These numbers are used to help fetch a record from the array C2:F11.

5. In cells A14:A23, enter integers 1–10, respectively, too. We are going to construct the proximity matrix in cells B14:K23.

6. In cell B14, enter the formula

```
=IFERROR(IF($A14=B$13,1000000, SQRT(SUMXMY2($C2:$F2,INDEX
($C$2:$F$11,B$13,0)))),1000000)
```

   In this formula, `INDEX($C$2:$F$11,B$13,0)` picks a record referenced by B13. In this particular case, it is the first data point (C2:F2).

   `SQRT(SUMXMY2($C2:$F2,INDEX($C$2:$F$11,B$13,0)))` computes the distance between C2:F2 and C2:F2.

Inside the IF function, the integer 1000000 is used as the maximum distance. If $A14=B$13, we know that it is computing the distance between the same data point (in this particular case, it is between C2:F2 and C2:F2). Though such a distance should be 0, but we would like to assign it the maximum distance for later use. Else the calculation result of the SQRT function will be returned.

The IFERROR function is used to make sure whenever there is an error, the maximum distance is assigned. This is important and necessary!

7. Autofill from B14 to K23. Now the proximity matrix has been constructed.

8. Enter the formula `=MATCH(MIN($B$14:$K$23),B14:K14,0)` in cell L14. Autofill from L14 to L23. The proximity matrix should look like Figure 8-11 and it tells that data points 4 and 1 have the shortest distance between.

		1	2	3	4	5	6	7	8	9	10	
14	1	1000000	37.88139	40.58325	17	57.77543	39.53479	160.8726	22.04541	70.47695	57.16642	4
15	2	37.88139	1000000	35.4683	52.64979	48.89785	65.31462	193.8324	47.86439	52.47857	89.02247	#N/A
16	3	40.58325	35.4683	1000000	55.29919	18.9473	56.76266	192.4968	31.89044	33.79349	92.8278	#N/A
17	4	17	52.64979	55.29919	1000000	71.72866	35.24202	144.7515	29.3428	86.56789	42.22558	1
18	5	57.77543	48.89785	18.9473	71.72866	1000000	65.55151	202.9631	47.01064	28.93095	106.0472	#N/A
19	6	39.53479	65.31462	56.76266	35.24202	65.55151	1000000	138.416	36.51027	89.72179	45.83667	#N/A
20	7	160.8726	193.8324	192.4968	144.7515	202.9631	138.416	1000000	163.3218	225.5549	106.0377	#N/A
21	8	22.04541	47.86439	31.89044	29.3428	47.01064	36.51027	163.3218	1000000	63.58459	66.10598	#N/A
22	9	70.47695	52.47857	33.79349	86.56789	28.93095	89.72179	225.5549	63.58459	1000000	125.1119	#N/A
23	10	57.16642	89.02247	92.8278	42.22558	106.0472	45.83667	106.0377	66.10598	125.1119	1000000	#N/A

Figure 8-11. *The proximity matrix and the first cluster*

9. Step-2:

   - Now we need to update the cluster information for the ten data points. Copy the numbers inside G2:G11 to H2:H11.
   - Update the number inside H2 and H5 to 1. Here, 1 indicates this is the first cluster.

- Since countries L1 and L4 are inside cluster 1, we need to compute the centroid of cluster 1. In cell N14, enter the formula `=AVERAGE(C2, C5)` and then autofill to Q14. Be advised that a cluster is represented by its centroid in this exercise.
- Copy the content of N14:Q14 to N15:Q15 by values only. Note, it is very important to copy N14:Q14 and paste by values to N15:Q15. The purpose is to avoid accidental iterative calculation in next step.
- Copy N15:Q15 to C2:F2, and then clear the content in C5:F5. Our worksheet should look like Figure 8-12.

  Note, it is very important to clear the content in C5:F5 for country L4 because country L1 (C2:F2) is now representing cluster 1.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Fixed	Country	Murder	Assault	UrbanPop	Rape	Step-1	Step-2	Step-3	Step-4	Step-5	Step-6	Step-7	Step-8	Step-9	Step-10
2	1	L1	10	185	50.5	19	11	1								
3	2	L2	9	222	39	42	12	12								
4	3	L3	6	226	72	30	13	13								
5	4	L4					14	1								
6	5	L5	7	235	86	39	15	15								
7	6	L6	5	170	78	37	16	16								
8	7	L7	0	35	75	7	17	17								
9	8	L8	4	198	70	15	18	18								
10	9	L9	13	259	74	30	19	19								
11	10	L10	15	136	52	24	20	20								
12																
13		1	2	3	4	5	6	7	8	9	10					
14	1	1000000	45.06939	47.75196	1000000	64.56973	36.47259	152.7883	24.5204	78.47452	49.53029	#N/A		10	185	50.5
15	2	45.06939	1000000	35.4683	1000000	48.89785	65.31462	193.8324	47.86439	52.47857	89.02247	#N/A		10	185	50.5
16	3	47.75196	35.4683	1000000	1000000	18.9473	56.76266	192.4968	31.89044	33.79349	92.8278	5				
17	4	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A				
18	5	64.56973	48.89785	18.9473	1000000	1000000	65.55151	202.9631	47.01064	28.93095	106.0472	3				
19	6	36.47259	65.31462	56.76266	1000000	65.55151	1000000	138.416	36.51027	89.72179	45.83667	#N/A				
20	7	152.7883	193.8324	192.4968	1000000	202.9631	138.416	1000000	163.3218	225.5549	106.0377	#N/A				
21	8	24.5204	47.86439	31.89044	1000000	47.01064	36.51027	163.3218	1000000	63.58459	66.10598	#N/A				
22	9	78.47452	52.47857	33.79349	1000000	28.93095	89.72179	225.5549	63.58459	1000000	125.1119	#N/A				
23	10	49.53029	89.02247	92.8278	1000000	106.0472	45.83667	106.0377	66.10598	125.1119	1000000	#N/A				

Figure 8-12. *After step-2, the cell array L14:L23 automatically shows that countries L5 and L3 are in a cluster together*

10. Step-3:

    - Figure 8-12 shows that countries L3 and L5 are in a cluster. Since the current cluster values of L3 and L5 are 13 and 15, both greater than 10, we know that L3 and L5 haven't be clustered. Thus, they should be in cluster 2 together.

- Copy H2:H11 to I2:I11.
- Update cells I4 and I6 to 2.
- Time to compute the centroid of countries L3 and L5. In cell N14, enter `=AVERAGE(C4,C6)` and then autofill from N14 to Q14.
- Copy N14:Q14 and paste them by values into N15:Q15.
- Copy and paste N15:Q15 into C4:F4. We are using C4:F4 (country L3) to represent the cluster 2.
- Clear the content in C6:F6.

  Our worksheet should look like Figure 8-13.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Fixed	Country	Murder	Assault	UrbanPop	Rape	Step-1	Step-2	Step-3	Step-4	Step-5	Step-6	Step-7	Step-8	Step-9	Step-10	
2	1	L1	10	185	50.5	19	11	1	1								
3	2	L2	9	222	39	42	12	12	12								
4	3	L3	6.5	230.5	79	34.5	13	13	2								
5	4	L4					14	1	1								
6	5	L5					15	15	2								
7	6	L6	5	170	78	37	16	16	16								
8	7	L7	0	35	75	7	17	17	17								
9	8	L8	4	198	70	15	18	18	18								
10	9	L9	13	259	74	30	19	19	19								
11	10	L10	15	136	52	24	20	20	20								
12																	
13		1	2	3	4	5	6	7	8	9	10						
14	1	1000000	45.06939	55.99107	1000000	1000000	36.47259	152.7883	24.5204	78.47452	49.53029	8		6.5	230.5	79	34.5
15	2	45.06939	1000000	41.65033	1000000	1000000	65.31462	193.8324	47.86439	52.47857	89.02247	#N/A		6.5	230.5	79	34.5
16	3	55.99107	41.65033	1000000	1000000	1000000	60.57846	197.5721	39.03524	29.99583	99.20559	#N/A					
17	4	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A					
18	5	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A					
19	6	36.47259	65.31462	60.57846	1000000	1000000	1000000	138.416	36.51027	89.72179	45.83667	#N/A					
20	7	152.7883	193.8324	197.5721	1000000	1000000	138.416	1000000	163.3218	225.5549	106.0377	#N/A					
21	8	24.5204	47.86439	39.03524	1000000	1000000	36.51027	163.3218	1000000	63.58459	66.10598	1					
22	9	78.47452	52.47857	29.99583	1000000	1000000	89.72179	225.5549	63.58459	1000000	125.1119	#N/A					
23	10	49.53029	89.02247	99.20559	1000000	1000000	45.83667	106.0377	66.10598	125.1119	1000000	#N/A					

Figure 8-13. *Countries L3 and L5 are in cluster 2, Country L8 is in the same cluster as L1*

11. Step-4:
    - Figure 8-13 also automatically shows that country L8 is in the same cluster as L1. As I2:I11 of Step-3 reveals that L8 has not been clustered (its current cluster value is 18) but L1 has, L8 should now be assigned to cluster 1 with L1.
    - Copy I2:I11 to J2:J11; update J9 to 1.

- Recompute the centroid of cluster 1. Enter `=SUM(C2*2,C9)/3` in cell N14; autofill from N14 to Q14. Note, because cluster 1 already has two data points and now includes country L8, we need to double C2 first.
- Copy N14:Q14 and paste by values to N15:Q15, then copy N15:Q15 to C2:F2.
- Clear the content inside C9:F9. Our worksheet should disclose that countries L3 and L9 are in the same cluster. Refer to Figure 8-14.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Fixed	Country	Murder	Assault	UrbanPop	Rape	Step-1	Step-2	Step-3	Step-4	Step-5	Step-6
2	1	L1	8	189.3333	57	17.66667	11	1	1	1		
3	2	L2	9	222	39	42	12	12	12	12		
4	3	L3	6.5	230.5	79	34.5	13	13	2	2		
5	4	L4					14	1	1	1		
6	5	L5					15	15	2	2		
7	6	L6	5	170	78	37	16	16	16	16		
8	7	L7	0	35	75	7	17	17	17	17		
9	8	L8					18	18	18	1		
10	9	L9	13	259	74	30	19	19	19	19		
11	10	L10	15	136	52	24	20	20	20	20		
12												
13		1	2	3	4	5	6	7	8	9	10	
14	1	1000000	44.54461	49.64177	1000000	1000000	34.60572	155.9505	1000000	72.93528	54.39261	#N/A
15	2	44.54461	1000000	41.65033	1000000	1000000	65.31462	193.8324	1000000	52.47857	89.02247	#N/A
16	3	49.64177	41.65033	1000000	1000000	1000000	60.57846	197.5721	1000000	29.99583	99.20559	→ 9
17	4	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A
18	5	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A
19	6	34.60572	65.31462	60.57846	1000000	1000000	1000000	138.416	1000000	89.72179	45.83667	#N/A
20	7	155.9505	193.8324	197.5721	1000000	1000000	138.416	1000000	1000000	225.5549	106.0377	#N/A
21	8	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A
22	9	72.93528	52.47857	29.99583	1000000	1000000	89.72179	225.5549	1000000	1000000	125.1119	→ 3
23	10	54.39261	89.02247	99.20559	1000000	1000000	45.83667	106.0377	1000000	125.1119	1000000	#N/A

Figure 8-14. *After Step-4, countries L3 and L9 are in the same cluster*

12. Step-5:
    - Certainly, country L9 should be in cluster 2 with L3. Copy J2:J11 to K2:K11, and update K10 to 2.
    - To recompute the centroid of cluster 2, enter `=SUM(C4*2, C10)/3` in N14; autofill from N14 to Q14.

- Copy N14:Q14 and paste by values into N15:Q15.
- Copy and paste N15:Q15 to C4:F4, then clear the content of C10:F10. The proximity matrix will reveal that country L6 is in the same cluster with L1.

13. Step-6:
    - Copy K2:K11 to L2:L11; update cell L7 (note, it is the cell L7 not the Country L7!) to 1 to mark that country L6 is in cluster 1.
    - To recompute the centroid for cluster 1, enter `=SUM(C2*3, C7)/4` in N14; autofill from N14 to Q14.
    - Copy/paste from N14:Q14 by values to N15:Q15, then copy/paste N15:Q15 to C2:F2.
    - Clear the content inside C7:F7.

    Our worksheet should look like Figure 8-15.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Fixed	Country	Murder	Assault	UrbanPop	Rape	Step-1	Step-2	Step-3	Step-4	Step-5	Step-6
2	1	L1	7.25	184.5	62.25	22.5	11	1	1	1	1	1
3	2	L2	9	222	39	42	12	12	12	12	12	12
4	3	L3	8.666667	240	77.33333	33	13	13	2	2	2	2
5	4	L4					14	1	1	1	1	1
6	5	L5					15	15	2	2	2	2
7	6	L6					16	16	16	16	16	1
8	7	L7	0	35	75	7	17	17	17	17	17	17
9	8	L8					18	18	18	1	1	1
10	9	L9					19	19	19	19	2	2
11	10	L10	15	136	52	24	20	20	20	20	20	20
12												
13		1	2	3	4	5	6	7	8	9	10	
14	1	1000000	48.27137	58.48088	1000000	1000000	1000000	151.0153	1000000	1000000	50.19587	#N/A
15	2	48.27137	1000000	43.29614	1000000	1000000	1000000	193.8324	1000000	1000000	89.02247	3
16	3	58.48088	43.29614	1000000	1000000	1000000	1000000	206.837	1000000	1000000	107.6052	2
17	4	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A
18	5	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A
19	6	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A
20	7	151.0153	193.8324	206.837	1000000	1000000	1000000	1000000	1000000	1000000	106.0377	#N/A
21	8	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A
22	9	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A
23	10	50.19587	89.02247	107.6052	1000000	1000000	1000000	106.0377	1000000	1000000	1000000	#N/A

Figure 8-15. *After Step-6, countries L2 and L3 are in the same cluster*

14. Step-7:
    - Copy the cell array L2:L11 to M2:M11.
    - Since country L2 is in the same cluster with L3, it must be in cluster 2. Update M3 to 2.
    - To recompute the centroid of cluster 2, enter `=SUM(C4*3,C3)/4` in N14; autofill from N14 to Q14.
    - Again, we need to copy N14:Q14 and paste by values into N15:Q15, then copy N15:Q15 into C4:F4.
    - Clear the content of C3:F3.
15. Step-8:
    - At this point, the proximity matrix reveals that countries L1 and L10 are in the same cluster. Let's copy M2:M11 to N2:N11 and update N11 to 1.
    - In N14, enter the formula `=SUM(C2*4,C11)/5` and autofill from N14 to Q14.
    - Copy/paste N14:Q14 by values to N15:Q15, then copy N15:Q15 to C2:F2.
    - After clearing the content in C11:F11, the proximity matrix shows that countries L3 and L1 are in the same cluster. This is demonstrated in Figure 8-16.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Fixed	Country	Murder	Assault	UrbanPop	Rape	Step-1	Step-2	Step-3	Step-4	Step-5	Step-6	Step-7	Step-8
2	1	L1	8.8	174.8	60.2	22.8	11	1	1	1	1	1	1	1
3	2	L2					12	12	12	12	12	12	2	2
4	3	L3	8.75	235.5	67.75	35.25	13	13	2	2	2	2	2	2
5	4	L4					14	1	1	1	1	1	1	1
6	5	L5					15	15	2	2	2	2	2	2
7	6	L6					16	16	16	16	16	1	1	1
8	7	L7	0	35	75	7	17	17	17	17	17	17	17	17
9	8	L8					18	18	18	1	1	1	1	1
10	9	L9					19	19	19	19	2	2	2	2
11	10	L10					20	20	20	20	20	20	20	1
12														
13		1	2	3	4	5	6	7	8	9	10			
14	1	1000000	1000000	62.42193	1000000	1000000	1000000	141.7398	1000000	1000000	1000000	→ 3		7.04
15	2	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A		8.8
16	3	62.42193	1000000	1000000	1000000	1000000	1000000	202.799	1000000	1000000	1000000	→ 1		
17	4	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A		
18	5	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A		
19	6	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A		
20	7	141.7398	1000000	202.799	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A		
21	8	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A		
22	9	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A		
23	10	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A		

Figure 8-16. *After Step-8, countries L1 and L3 are inside the same cluster*

16. Step-9:

    - Since country L1 is in cluster 1 and L3 is in cluster 2, both have been processed. This indicates that we need to merge cluster 1 and cluster 2 now. Thus, copy N2:N11 to O2:O11. Pay attention that there are five 1s and four 2s in O2:O11.
    - Do not update O2:O11 at this moment until we have finished recomputing the centroid of the merged clusters. Enter `=SUM(C2*5, C4*4)/9` in N14 and autofill from N14 to Q14.
    - Copy N14:Q14 and paste by values into N15:Q15.
    - Now update all the 2s inside O2:O11 to 1.
    - Make sure that the values of N15:Q15 are copied to C2:F2, then clear the content of C4:F4.

17. Step-10:

    - At this moment, the proximity matrix should tell that L7 is in the same cluster with L1.
    - Thus, we can fill in 1s in P2:P11.

Our worksheet should look the same as Figure 8-17.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Fixed	Country	Murder	Assault	UrbanPop	Rape	Step-1	Step-2	Step-3	Step-4	Step-5	Step-6	Step-7	Step-8	Step-9	Step-10
2	1	L1	8.777778	201.7778	63.55556	28.33333	11	1	1	1	1	1	1	1	1	1
3	2	L2					12	12	12	12	12	12	2	2	1	1
4	3	L3					13	13	2	2	2	2	2	2	1	1
5	4	L4					14	1	1	1	1	1	1	1	1	1
6	5	L5					15	15	2	2	2	2	2	2	1	1
7	6	L6					16	16	16	16	16	1	1	1	1	1
8	7	L7	0	35	75	7	17	17	17	17	17	17	17	17	17	1
9	8	L8					18	18	18	1	1	1	1	1	1	1
10	9	L9					19	19	19	19	2	2	2	2	1	1
11	10	L10					20	20	20	20	20	20	20	1	1	1
12																
13		1	2	3	4	5	6	7	8	9	10					
14	1	1000000	1000000	1000000	1000000	1000000	1000000	168.7541	1000000	1000000	1000000	7		4.8765	112.1	35.309
15	2	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A		8.777778	201.7778	63.55556
16	3	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A				
17	4	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A				
18	5	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A				
19	6	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A				
20	7	168.7541	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1				
21	8	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A				
22	9	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A				
23	10	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	1000000	#N/A				

Figure 8-17. *All data points are clustered into cluster 1 at Step-10*

18. Our task stays unfinished. We need to organize the ten data points so that we can have a better idea how to draw the dendrogram. To do so, we need to sort the table B1:P11.

    Select B1:P11 ➤ click Data ➤ Sort ➤ at "Sort by", select Step-2 and then click OK. This is explained in Figure 8-18. Note, "My data has headers" is checked.

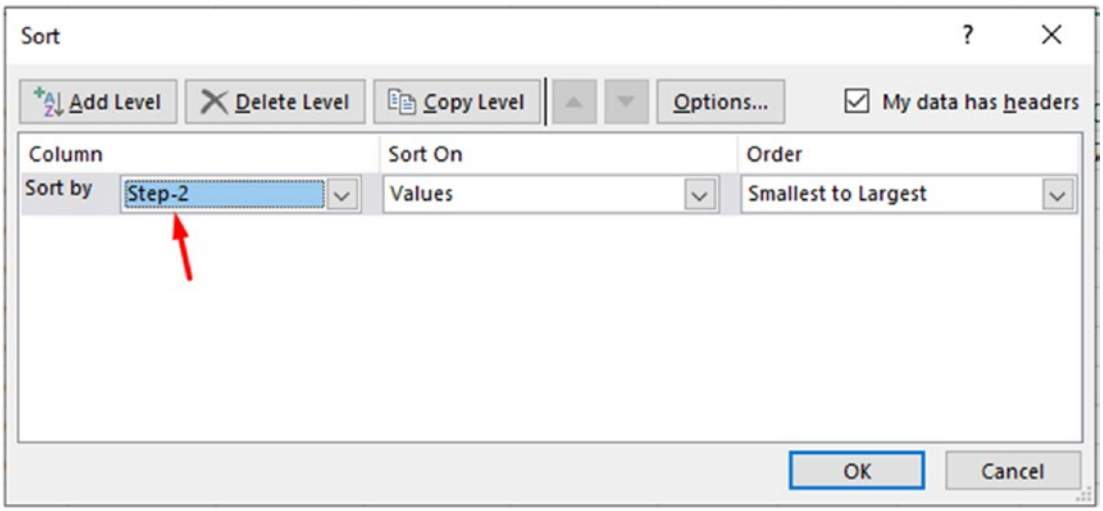


Figure 8-18. *Sort B1:P11 by Step-2. Note, "My data has headers" is checked*

19. At "Sort by", change Step-2 to Step-3 to sort B1:P11 by Step-3. Don't forget to click OK to complete the sorting process.

20. Continue sorting by Step-4, …, and Step-9, respectively, and in order. Part of our worksheet should look like Figure 8-19.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Fixed	Country	Murder	Assault	UrbanPop	Rape	Step-1	Step-2	Step-3	Step-4	Step-5	Step-6	Step-7	Step-8	Step-9	Step-10
2	1	L1	8.777778	201.7778	63.55556	28.33333	11	1	1	1	1	1	1	1	1	1
3	2	L4					14	1	1	1	1	1	1	1	1	1
4	3	L8					18	18	18	1	1	1	1	1	1	1
5	4	L6					16	16	16	16	16	1	1	1	1	1
6	5	L10					20	20	20	20	20	20	20	1	1	1
7	6	L3					13	13	2	2	2	2	2	2	1	1
8	7	L5					15	15	2	2	2	2	2	2	1	1
9	8	L9					19	19	19	19	2	2	2	2	1	1
10	9	L2					12	12	12	12	12	12	2	2	1	1
11	10	L7	0	35	75	7	17	17	17	17	17	17	17	17	17	1

Figure 8-19. *After sorting, the data is ready for drawing a dendrogram*

21. After sorting, the data is ready for drawing a dendrogram. Take a close look at the cell array B1:B11, which shows that when we draw our dendrogram, we should arrange our data points (countries) from left to right in this order: L1, L4, L8, L6, L10, L3, L5, L9, L2, L7.

    Data inside G1:P11 instruct us the following:

    - (L1, L4) is the first level of cluster.
    - (L3, L5) is the second level of cluster.
    - ((L1, L4), L8) is the third level of cluster.
    - ((L3, L5), L9) is the fourth level of cluster.
    - (((L1, L4), L8), L6) is the fifth level of cluster, and so on.

    The current result without the dendrogram can be found in Chapter8-2b.xlsx. Please try to apply what you have learned regarding drawing a dendrogram to draw a dendrogram inside Chapter8-2b.xlsx.

    Note, I completed the hierarchical clustering process inside the same worksheet. Should you want to keep the intermediate steps intact, you can make a copy of the worksheet at each of the intermediate steps, similar to what we did in k-means clustering. In fact, keeping intermediate steps intact can help avoid errors.

Chapter8-2e.xlsx repeats the preceding process by copying worksheets to keep intermediate steps intact. Please take a look if you are interested.

22. To draw the dendrogram, it is a good idea to delineate it on a piece of paper with a pen first (or on an iPad-like device). This will greatly help us specify the X and Y values for the upcoming scatter plot. I won't go over in detail how the X and Y values are generated, but they are available in Chapter8-2c.xlsx which you should open now.

23. In Chapter8-2c.xlsx, the X and Y values for our dendrogram is presented in cell array R1:S54. Please select them and draw a "X Y Scatter with Straight Lines". We shall get a chart like Figure 8-20.

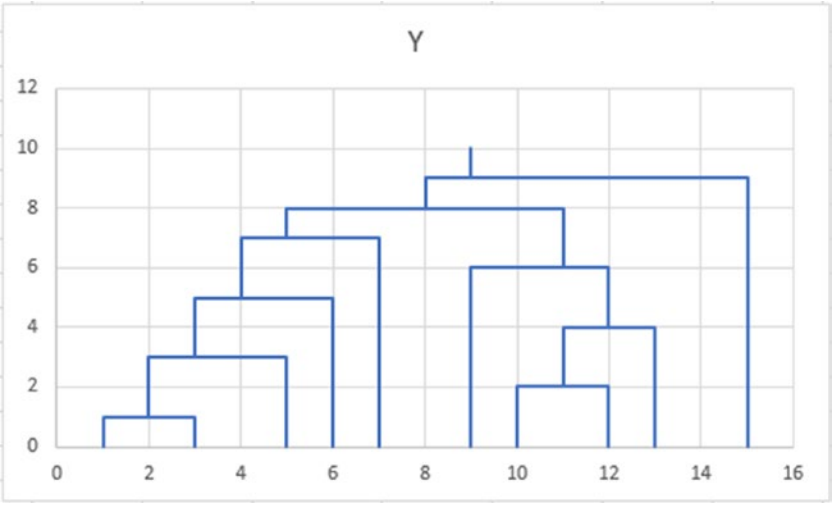


Figure 8-20. *The X Y Scatter with Straight Lines for our dendrogram*

24. Please revise the title to "Dendrogram".

25. Click on the x-axis to delete it.

26. Click "Add Chart Elements" ➤ Axis Titles.

27. Revise the y-axis title to "Distance not in proportion". As indicated by this y-axis title, the clade heights in our dendrogram do not reflect the proportional relationship of the cluster distances.

28. Revise the x-axis title to "L1 L4 L8 L6 L10 L3 L5 L9 L2 L7". After that, add or remove spaces between the words so that they align well with your chart.

    This approach is different from applying data label. Certainly, you can try to add data labels instead. Figure 8-21 shows the dendrogram I created by adjusting x-axis title content.

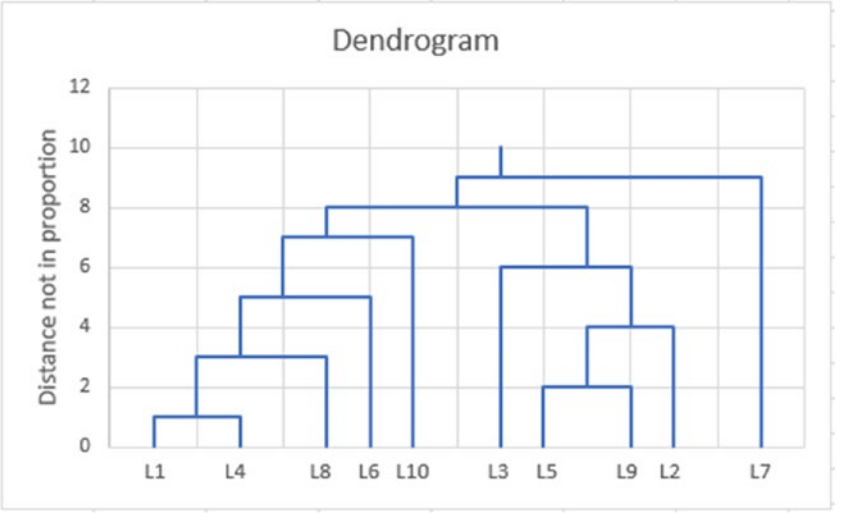


Figure 8-21. *One final dendrogram. Observe that the distances between the text words inside the x-axis title are explicitly enlarged*

The final result can be found in Chapter8-2d.xlsx which also draws a dendrogram by applying data labels.

Reinforcement Exercises

You may have realized that accomplishing hierarchical clustering and dendrogram via Excel is error prone because of the necessary manual work. This is especially true when the number of data points increases. As I mentioned before, one way to mitigate errors is to keep intermediate steps intact by making copies of the working worksheet, similar to what we did in k-means clustering.

Can we conduct hierarchical clustering on hundreds of data points through Excel? Certainly we can. But we must borrow a hand from k-means clustering first. The idea is to apply k-means clustering (e.g., k=7) on the larger dataset first, then treat each cluster as one data point to start hierarchical clustering.

- The file Chapter8-HW-1.xlsx provides an opportunity for you to practice hierarchical clustering and dendrogram drawing. Refer to Chapter8-HW-1-withAnswers.xlsx whenever you need some help.
- Chapter8-HW-2.xlsx allows you to practice hierarchical clustering with a larger dataset such that you need to apply k-means clustering first. Chapter8-HW-2-withAnswers.xlsx shows how to achieve that. Chapter8-HW-2-withAnswers.xlsx also shows the hierarchical clustering process by keeping intermediate steps intact.
- Note, Chapter8-2e.xlsx also shows the hierarchical clustering process by making copies of worksheets to keep intermediate steps intact. I strongly suggest that you take a close look at this file.

Review Points

For Chapter 8, the following knowledge, skills, and functions are worth reviewing:

1. Hierarchical clustering and dendrogram
2. The algorithm for agglomerative clustering
3. Proximity matrix
4. The Excel built-in features SORT and Paste by Values
5. Scatter plot chart and data labels
6. Functions IF, MIN, IFERROR, INDEX, MATCH, SQRT, and SUMXMY2
7. Copy worksheet to keep intermediate steps intact